# Geometry, Triangles & Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • [FRQ] • Target: Parallel Lines & Transversal Angle Relationships

Two parallel lines $\ell_1 \parallel \ell_2$ are intersected by a transversal $t$. If a pair of consecutive interior (co-interior) angles have measures $(3x + 15)^\circ$ and $(2x - 5)^\circ$, find $x$ and the measure of both angles.

Consecutive interior angles are supplementary: $(3x + 15) + (2x - 5) = 180 \Rightarrow 5x + 10 = 180 \Rightarrow 5x = 170 \Rightarrow x = 34$. The two angles measure $3(34) + 15 = 117^\circ$ and $2(34) - 5 = 63^\circ$.

**2** [Foundational] • [FRQ] • Target: Triangle Angle-Sum & Exterior Angle Theorem

In $\Delta ABC$, the exterior angle at vertex $C$ measures $134^\circ$, and the interior angle $\angle A$ measures $58^\circ$. Find $\angle B$ and $\angle ACB$.

By the Exterior Angle Theorem, $\angle A + \angle B = 134^\circ \Rightarrow 58^\circ + \angle B = 134^\circ \Rightarrow \angle B = 76^\circ$. The interior angle at $C$ is $\angle ACB = 180^\circ - 134^\circ = 46^\circ$.

**3** [Foundational] • [FRQ] • Target: Pythagorean Theorem & Right-Triangle Trig Ratios

In right triangle $ABC$ shown below, $\angle C = 90^\circ$, $AC = 15$, and $BC = 8$. Find (a) the hypotenuse $AB$, (b) $\sin A$, (c) $\cos A$, and (d) $\tan B$.

(a) By the Pythagorean theorem, $AB = \sqrt{15^2 + 8^2} = \sqrt{289} = 17$. (b) $\sin A = \frac{8}{17}$. (c) $\cos A = \frac{15}{17}$. (d) $\tan B = \frac{15}{8}$.

**4** [Foundational] • [FRQ] • Target: Special Right Triangles ($30^{\circ}\mathrm{-}60^{\circ}\mathrm{-}90^{\circ}$)

Find the exact perimeter and area of an equilateral triangle whose altitude has length $6\sqrt{3}$ cm.

In a $30^\circ - 60^\circ - 90^\circ$ half-triangle, altitude $h = \frac{\sqrt{3}}{2}s = 6\sqrt{3} \Rightarrow s = 12$ cm. Thus the perimeter is $3s = 36$ cm and the area is $\frac{1}{2}(12)(6\sqrt{3}) = 36\sqrt{3}$ cm$^2$.

**5** [Foundational] • [FRQ] • Target: Special Right Triangles ($45^{\circ}\mathrm{-}45^{\circ}\mathrm{-}90^{\circ}$)

A square has diagonal of length $10\sqrt{2}$ m. Find its side length, perimeter, and area.

In a $45^\circ - 45^\circ - 90^\circ$ right triangle, $d = s\sqrt{2} = 10\sqrt{2} \Rightarrow s = 10$ m. Perimeter $= 4s = 40$ m, and Area $= s^2 = 100$ m$^2$.

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**6** [Foundational] • [FRQ] • Target: Triangle Inequality Theorem

Two sides of a triangle have lengths $7$ and $12$. State the strict inequality for all possible lengths $x$ of the third side, and find how many integer values $x$ can take.

By the Triangle Inequality Theorem, $|12-7| < x < 12+7 \Rightarrow 5 < x < 19$. The possible integer lengths are $6, 7, \ldots, 18$, which gives $18-6+1 = 13$ integer values.

**7** [Foundational] • [FRQ] • Target: Isosceles Triangle Properties

In $\Delta PQR$, $PQ = PR$ (isosceles) and the vertex angle $\angle P$ measures $44^\circ$. Find the measure of each base angle $\angle Q$ and $\angle R$.

By the Isosceles Triangle Theorem, $\angle Q = \angle R = \frac{180^\circ - 44^\circ}{2} = \frac{136^\circ}{2} = 68^\circ$.

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## Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

**8** [Intermediate] • [FRQ] • Target: Similar Triangles & Parallel Cross-Sections

In $\Delta ABC$, segment $DE$ is parallel to base $BC$ with $D$ on $AB$ and $E$ on $AC$. Given $AD = 6$, $DB = 9$, and $DE = 8$, find the length of $BC$ and the ratio of the area of $\Delta ADE$ to the area of $\Delta ABC$.

Since $DE \parallel BC$, $\Delta ADE \sim \Delta ABC$ with scale factor $k = \frac{AD}{AB} = \frac{6}{6+9} = \frac{6}{15} = \frac{2}{5}$. Thus $\frac{8}{BC} = \frac{2}{5} \Rightarrow BC = 20$. The area ratio is $k^2 = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$.

**9** [Intermediate] • [FRQ] • Target: Complementary Angle Identity $\sin\theta = \cos(90^{\circ}-\theta)$

In right triangle $DEF$ with $\angle F = 90^\circ$, $\sin D = \frac{5}{13}$. Find $\cos E$, $\cos D$, and $\tan E$.

Since $D$ and $E$ are complementary ($D + E = 90^\circ$), $\cos E = \sin D = \frac{5}{13}$. By the Pythagorean identity, $\cos D = \sqrt{1 - (5/13)^2} = \frac{12}{13}$, and $\tan E = \cot D = \frac{12}{5}$.

**10** [Intermediate] • [FRQ] • Target: Geometric Mean Theorems in Right Triangles

In right triangle $ABC$, the altitude $CD$ is drawn to the hypotenuse $AB$. If $AD = 4$ and $DB = 9$, find the length of the altitude $CD$ and both legs $AC$ and $BC$.

By the Geometric Mean (Right-Triangle Altitude) Theorem: $CD = \sqrt{AD \cdot DB} = \sqrt{4 \cdot 9} = 6$. Since $AB = 4 + 9 = 13$, $AC = \sqrt{AD \cdot AB} = \sqrt{4 \cdot 13} = 2\sqrt{13}$ and $BC = \sqrt{DB \cdot AB} = \sqrt{9 \cdot 13} = 3\sqrt{13}$.

**11** [Intermediate] • [FRQ] • Target: Interior & Exterior Angles of Regular Polygons

Find the measure of one interior angle and one exterior angle of a regular $15$-sided polygon (pentadecagon).

Each exterior angle of a regular $n$-gon measures $\frac{360^\circ}{n} = \frac{360^\circ}{15} = 24^\circ$. Therefore each interior angle measures $180^\circ - 24^\circ = 156^\circ$.

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**12** [Intermediate] • [FRQ] • Target: Angles of Elevation & Depression

A surveyor stands $40\,\text{m}$ from the base of a vertical radio tower on level ground. The angle of elevation to the top of the tower is $60^\circ$, and the surveyor's eye level is $1.6\,\text{m}$ above the ground. Find the exact height of the tower and its decimal approximation to two decimal places.

Height above eye level $= 40\tan 60^\circ = 40\sqrt{3}$ m. Adding eye height gives $H = 40\sqrt{3} + 1.6 \approx 69.28 + 1.6 = 70.88\,\text{m}$.

**13** [Intermediate] • [FRQ] • Target: Triangle Angle Bisector Theorem

In $\Delta ABC$, the angle bisector of $\angle A$ meets side $BC$ at point $D$. If $AB = 10$, $AC = 15$, and $BC = 20$, find the exact lengths of $BD$ and $DC$.

By the Angle Bisector Theorem, $\frac{BD}{DC} = \frac{AB}{AC} = \frac{10}{15} = \frac{2}{3}$. Since $BD + DC = 20$, we have $BD = \frac{2}{5}(20) = 8$ and $DC = \frac{3}{5}(20) = 12$.

**14** [Intermediate] • [FRQ] • Target: Isosceles Trapezoids & Pythagorean Decomposition

An isosceles trapezoid has parallel bases of lengths $10\,\text{cm}$ and $26\,\text{cm}$ and non-parallel legs of length $17\,\text{cm}$. Find the height $h$, the area, and the length of each diagonal of the trapezoid.

Dropping altitudes from the shorter base splits the longer base into $8 + 10 + 8$. By the Pythagorean theorem, $h = \sqrt{17^2 - 8^2} = 15\,\text{cm}$. Area $= \frac{10 + 26}{2} \times 15 = 270\,\text{cm}^2$. Each diagonal is the hypotenuse of a right triangle with legs $10 + 8 = 18$ and $15$: $d = \sqrt{18^2 + 15^2} = \sqrt{549} = 3\sqrt{61}$ cm.

## Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • [FRQ] • Target: Coordinate Triangle Geometry & Centroid • Cross-Lesson: Topic 1.01 (Coordinate Geometry)

In $\Delta ABC$, the vertices are $A(1, 2)$, $B(9, 2)$, and $C(5, 8)$. Verify that $\Delta ABC$ is isosceles, find the length of the altitude from $C$ to $AB$, and determine the coordinates of the centroid $G$.

Base $AB$ is horizontal of length $9 - 1 = 8$. Side $AC = \sqrt{(5-1)^2 + (8-2)^2} = \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$ and $BC = \sqrt{(5-9)^2 + (8-2)^2} = 2\sqrt{13}$, so $AC = BC$. The altitude from $C(5, 8)$ to $y = 2$ has length $8 - 2 = 6$. The centroid is $G = \left(\frac{1+9+5}{3}, \frac{2+2+8}{3}\right) = (5, 4)$.

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**16** **[Challenging] • [FRQ] • Target: Algebraic Right Triangles via Quadratics • Cross-Lesson: Topic 3.01 (Quadratic Equations)**

In a right triangle, one leg is $7\,\text{cm}$ longer than the other leg, and the hypotenuse is $17\,\text{cm}$. Set up and solve a quadratic equation to find the lengths of the two legs and the sine of the smaller acute angle.

Let the shorter leg be $x$ and the longer leg be $x+7$: $x^2+(x+7)^2=17^2 \Rightarrow 2x^2+14x+49=289 \Rightarrow x^2+7x-120=0 \Rightarrow (x+15)(x-8)=0$. Since $x>0$, the legs are $8\,\text{cm}$ and $15\,\text{cm}$, and the sine of the smaller acute angle (opposite the $8\,\text{cm}$ leg) is $\frac{8}{17}$.

**17** **[Challenging] • [FRQ] • Target: Law of Cosines & Trigonometric Triangle Area • Cross-Lesson: Topic 9.01 (Trigonometry)**

In $\Delta ABC$, $AB=7$, $BC=8$, and $\angle B=60^\circ$. Use the Law of Cosines to find the exact length of side $AC$ and the exact area of $\Delta ABC$.

By the Law of Cosines, $AC^2=7^2+8^2-2(7)(8)\cos 60^\circ=49+64-56=57 \Rightarrow AC=\sqrt{57}$. The area is $\frac{1}{2}(7)(8)\sin 60^\circ=28\cdot\frac{\sqrt{3}}{2}=14\sqrt{3}$.

**18** **[Challenging] • [FRQ] • Target: Thales' Theorem & Inscribed Right Triangles • Cross-Lesson: Topic 9.01 (Circles & Inscribed Angles)**

Triangle $ABC$ is inscribed in a circle of diameter $AB=20\,\text{cm}$. If $\cos A=\frac{3}{5}$, find the lengths of chords $AC$ and $BC$ and the area of the region inside the circle but outside $\Delta ABC$.

By Thales' Theorem, $\angle C=90^\circ$ because $AB$ is a diameter. Thus $AC=AB\cos A=20\left(\frac{3}{5}\right)=12\,\text{cm}$ and $BC=\sqrt{20^2-12^2}=16\,\text{cm}$. Triangle area $=\frac{1}{2}(12)(16)=96\,\text{cm}^2$; circle area $=\pi(10^2)=100\pi\,\text{cm}^2$. Region area $=(100\pi-96)\,\text{cm}^2$.

**19** **[Challenging] • [FRQ] • Target: Inscribed Square via Similar Right Triangles • Cross-Lesson: Topic 5.01 (Proportions) & Topic 8.01 (Area)**

A square is inscribed inside a right triangle with legs $6$ and $12$ so that one vertex of the square coincides with the right-angle vertex of the triangle and the opposite vertex lies on the hypotenuse. Find the side length $s$ of the square.

Let the square have side $s$. The upper-right small right triangle has legs $6-s$ and $s$, and is similar to the $6$–$12$ right triangle: $\frac{6-s}{s}=\frac{6}{12}=\frac{1}{2} \Rightarrow 12-2s=s \Rightarrow 3s=12 \Rightarrow s=4$.

**20** **[Challenging] • [FRQ] • Target: Orthogonal Medians & Centroid Area Partition • Cross-Lesson: Topic 8.01 (Area Decomposition)**

In $\Delta ABC$, medians $AD$ and $BE$ are perpendicular to each other and have lengths $AD=9$ and $BE=12$. Using the centroid ratio $2:1$, find the length of side $AB$ and the area of $\Delta ABC$.

The centroid $G$ divides each median in a $2:1$ ratio, so $AG=6$, $GD=3$ and $BG=8$, $GE=4$. Since $AD\perp BE$ at $G$, in right triangle $AGB$ we have $AB=\sqrt{6^2+8^2}=10$. The area of orthogonal-diagonal quadrilateral $ABDE$ (or using $\Delta ABC=3\cdot\Delta AGB$): $[AGB]=\frac{1}{2}(6)(8)=24$, and since the three sub-triangles $AGB, BGC, CGA$ have equal area, $[ABC]=3(24)=72$.

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## Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

**21** [Foundational] • [MCQ] • Target: Triangle Angle-Sum Theorem

In $\Delta ABC$, $\angle A = 42^\circ$ and $\angle B = 73^\circ$. What is the measure of $\angle C$?

**a.** $55^\circ$ **b.** $65^\circ$ ✓

**c.** $75^\circ$ **d.** $115^\circ$

The interior angles of a triangle sum to $180^\circ$: $\angle C = 180^\circ - (42^\circ + 73^\circ) = 65^\circ$.

**22** [Foundational] • [MCQ] • Target: Pythagorean Triples

In a right triangle, the legs have lengths $9$ and $12$. What is the length of the hypotenuse?

**a.** 13 **b.** 15 ✓

**c.** 18 **d.** 21

By the Pythagorean theorem (a $3{-}4{-}5$ multiple scaled by $3$), $c = \sqrt{9^2 + 12^2} = \sqrt{225} = 15$.

**23** [Foundational] • [MCQ] • Target: Complementary Trig Identities

If $\sin(3x - 10)^\circ = \cos(2x + 25)^\circ$, where both angles are acute, what is the value of $x$?

**a.** 13 **b.** 15 ✓

**c.** 17 **d.** 21

By the cofunction identity, the two acute angles are complementary:

$(3x - 10) + (2x + 25) = 90 \Rightarrow 5x + 15 = 90 \Rightarrow 5x = 75 \Rightarrow x = 15$.

**24** [Foundational] • [MCQ] • Target: Polygon Interior Angle Sum

What is the sum of the interior angles of a convex octagon ($8$ sides)?

**a.** $900^\circ$ **b.** $1080^\circ$ ✓

**c.** $1260^\circ$ **d.** $1440^\circ$

Sum of interior angles $= (n - 2) \times 180^\circ = (8 - 2) \times 180^\circ = 1080^\circ$.

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**25** [Foundational] • [MCQ] • Target: Proportions in Similar Triangles

Triangles $ABC$ and $DEF$ are similar, with $AB$ corresponding to $DE$. If $AB = 6$, $BC = 10$, and $DE = 9$, what is the length of $EF$?

**a.** 12

**b.** 13

**c.** 15 ✓

**d.** 18

Set up the proportion $\frac{EF}{BC} = \frac{DE}{AB} \Rightarrow \frac{EF}{10} = \frac{9}{6} = 1.5 \Rightarrow EF = 15$.

**26** [Foundational] • [MCQ] • Target: Special Right Triangle Ratios

In a $30^\circ-60^\circ-90^\circ$ right triangle, the side opposite the $30^\circ$ angle has length $7$. What is the length of the side opposite the $60^\circ$ angle?

**a.** $7\sqrt{2}$

**b.** $7\sqrt{3}$ ✓

**c.** 14

**d.** $\frac{7\sqrt{3}}{2}$

In a $30^\circ-60^\circ-90^\circ$ triangle, the legs are in the ratio $1 : \sqrt{3}$, so the side opposite $60^\circ$ is $7\sqrt{3}$.

**27** [Foundational] • [MCQ] • Target: Alternate Interior & Supplementary Angles

In the figure below, lines $\ell$ and $m$ are parallel. If $\angle 1 = 128^\circ$, what is the measure of its alternate interior angle's supplement?

**a.** $38^\circ$

**b.** $52^\circ$ ✓

**c.** $62^\circ$

**d.** $128^\circ$

The alternate interior angle also measures $128^\circ$, so its supplement measures $180^\circ - 128^\circ = 52^\circ$.

**28** [Foundational] • [MCQ] • Target: Testing the Triangle Inequality

Which of the following sets of side lengths CANNOT form a triangle?

**a.** $4, 5, 8$

**b.** $6, 6, 11$

**c.** $5, 9, 14$ ✓

**d.** $7, 10, 12$

In $5, 9, 14$, the sum of the two shorter sides is $5 + 9 = 14$, which fails the strict triangle inequality $a + b > c$.

## Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

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**29** [Intermediate] • [MCQ] • Target: Perimeter from Trigonometric Ratio

In right triangle $PQR$ shown below, $\angle R = 90^\circ$, $PR = 12$, and $\tan P = \frac{3}{4}$. What is the perimeter of $\Delta PQR$?

**a.** 28

**b.** 32

**c.** 36 ✓

**d.** 48

Since $\tan P = \frac{QR}{PR} = \frac{3}{4}$ and $PR = 12$, we get $QR = 9$. Then $PQ = \sqrt{12^2 + 9^2} = 15$, so the perimeter is $12 + 9 + 15 = 36$.

**30** [Intermediate] • [MCQ] • Target: Area-to-Perimeter Ratio in Similar Triangles

Two similar triangles have perimeters in the ratio $3 : 7$. If the area of the smaller triangle is $27\,\text{cm}^2$, what is the area of the larger triangle?

**a.** $63\,\text{cm}^2$

**b.** $108\,\text{cm}^2$

**c.** $147\,\text{cm}^2$ ✓

**d.** $189\,\text{cm}^2$

Area scales as the square of the linear (perimeter) ratio: $\frac{A_2}{27} = \left(\frac{7}{3}\right)^2 = \frac{49}{9} \Rightarrow A_2 = 3 \times 49 = 147\,\text{cm}^2$.

**31** [Intermediate] • [MCQ] • Target: Altitude & Area of Isosceles Triangles

In an isosceles triangle with two sides of length $13$ and base of length $10$, what is the area of the triangle?

**a.** 60 ✓

**b.** 65

**c.** 120

**d.** 130

The altitude to the base bisects the base into segments of length $5$, forming two $5{-}12{-}13$ right triangles with height $h = 12$. Area $= \frac{1}{2}(10)(12) = 60$.

**32** [Intermediate] • [MCQ] • Target: Finding Number of Sides from Interior Angle

If each interior angle of a regular polygon measures $165^\circ$, how many sides does the polygon have?

**a.** 18

**b.** 20

**c.** 24 ✓

**d.** 30

Each exterior angle measures $180^\circ - 165^\circ = 15^\circ$. Since exterior angles sum to $360^\circ$, $n = \frac{360^\circ}{15^\circ} = 24$ sides.

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**33** [Intermediate] • [MCQ] • Target: Geometric Mean Theorem for Right-Triangle Legs

In right triangle $ABC$ with $\angle C = 90^\circ$, altitude $CD$ is drawn to hypotenuse $AB$. If $AC = 10$ and $AD = 5$, what is the length of $AB$?

**a.** 15

**b.** 20 ✓

**c.** 25

**d.** $5\sqrt{5}$

By the Right-Triangle Leg Geometric Mean Theorem, $AC^2 = AD \cdot AB \Rightarrow 10^2 = 5 \cdot AB \Rightarrow AB = 20$.

**34** [Intermediate] • [MCQ] • Target: Sliding Ladder Pythagorean Word Problems

A 25-foot ladder leans against a vertical wall with its base 7 feet from the wall. If the top of the ladder slides down the wall by 4 feet, how many feet does the base of the ladder slide away from the wall?

**a.** 4

**b.** 7

**c.** 8 ✓

**d.** 15

Initially, the top is at height $\sqrt{25^2 - 7^2} = 24$ ft. After sliding down 4 ft to height 20 ft, the new base distance is $\sqrt{25^2 - 20^2} = 15$ ft. Thus the base slides out by $15 - 7 = 8$ ft.

**35** [Intermediate] • [MCQ] • Target: Midsegment Theorem & Area Partition

In $\Delta ABC$, $D$ is the midpoint of $AB$ and $E$ is the midpoint of $AC$. If the area of trapezoid $DBCE$ is 45, what is the area of $\Delta ABC$?

**a.** 54

**b.** 60 ✓

**c.** 75

**d.** 90

Midsegment $DE$ forms $\Delta ADE \sim \Delta ABC$ with ratio $1:2$, so $[ADE] = \frac{1}{4}[ABC]$ and trapezoid $[DBCE] = \frac{3}{4}[ABC] = 45 \Rightarrow [ABC] = 60$.

**36** [Intermediate] • [MCQ] • Target: Pythagorean Trigonometric Identities

If $\theta$ is an acute angle such that $\sin\theta = 3\cos\theta$, what is the value of $\sin\theta\cos\theta$?

**a.** $\frac{1}{10}$

**b.** $\frac{3}{10}$ ✓

**c.** $\frac{1}{3}$

**d.** $\frac{3}{5}$

Since $\tan\theta = 3$, a right triangle has opposite leg 3, adjacent leg 1, and hypotenuse $\sqrt{10}$. Thus $\sin\theta\cos\theta = \left(\frac{3}{\sqrt{10}}\right)\left(\frac{1}{\sqrt{10}}\right) = \frac{3}{10}$.

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**37** [Intermediate] • [MCQ] • Target: Altitude Decomposition of Oblique Triangles

In $\Delta ABC$, $\angle A = 30^\circ$, $\angle C = 45^\circ$, and $AB = 12$. What is the exact length of $BC$?

**a.** $6\sqrt{2}$ ✓

**b.** $6\sqrt{3}$

**c.** $6\sqrt{6}$

**d.** $12\sqrt{2}$

Drop altitude $BD$ to $AC$: in the $30^\circ{-}60^\circ{-}90^\circ$ triangle $ABD$, $BD = \frac{1}{2}(12) = 6$. In the $45^\circ{-}45^\circ{-}90^\circ$ triangle $CBD$, $BC = BD\sqrt{2} = 6\sqrt{2}$.

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## Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • [MCQ] • Target: Inradius of a Right Triangle • Cross-Lesson: Topic 9.01 (Circles & Tangent Segments)

What is the radius of the circle inscribed in a right triangle with legs $9$ and $40$?

**a.** $3.5$

**b.** $4$ ✓

**c.** $4.5$

**d.** $5$

The hypotenuse is $\sqrt{9^2 + 40^2} = \sqrt{81 + 1600} = 41$. The inradius of a right triangle is $r = \frac{a + b - c}{2} = \frac{9 + 40 - 41}{2} = 4$.

**39** [Challenging] • [MCQ] • Target: Slope as Tangent & Trigonometric Ratios of Lines • Cross-Lesson: Topic 1.01 (Linear Intercepts & Slope)

In the $xy$-plane, line $y = -\frac{3}{4}x + 6$ forms a right triangle with the positive coordinate axes. What is the value of $\sin\theta$, where $\theta$ is the acute angle the line makes with the $x$-axis?

**a.** $\frac{3}{5}$ ✓

**b.** $\frac{4}{5}$

**c.** $\frac{3}{4}$

**d.** $\frac{4}{3}$

The intercepts are $(8, 0)$ and $(0, 6)$, and the hypotenuse has length $\sqrt{8^2 + 6^2} = 10$. The side opposite the angle with the $x$-axis is $6$, so $\sin\theta = \frac{6}{10} = \frac{3}{5}$.

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**40** [Challenging] • [MCQ] • Target: Heron's Formula & 13-14-15 Triangle Altitude • Cross-Lesson: Topic 8.01 (Area of Triangles)

In $\Delta ABC$, sides $a, b, c$ have lengths $13, 14, 15$. What is the length of the altitude to the side of length $14$?

**a.** 10

**b.** 11

**c.** 12 ✓

**d.** 12.5

Decompose the base $14$ into $5 + 9$: both $5-12-13$ and $9-12-15$ share the common altitude $h = 12$ (or by Heron's formula, $s = 21$, Area $= \sqrt{21(8)(7)(6)} = 84 = \frac{1}{2}(14)h \Rightarrow h = 12$).

**41** [Challenging] • [MCQ] • Target: Algebraic Squaring with $\sin^2\theta + \cos^2\theta = 1$ • Cross-Lesson: Topic 3.01 (Quadratic Identities)

If $\sin\theta + \cos\theta = \frac{7}{5}$ for an acute angle $\theta$, what is the value of $\sin\theta\cos\theta$?

**a.** $\frac{12}{25}$ ✓

**b.** $\frac{24}{25}$

**c.** $\frac{7}{25}$

**d.** $\frac{12}{5}$

Square both sides: $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 1 + 2\sin\theta\cos\theta = \frac{49}{25} \Rightarrow 2\sin\theta\cos\theta = \frac{24}{25} \Rightarrow \sin\theta\cos\theta = \frac{12}{25}$.

**42** [Challenging] • [MCQ] • Target: Isoperimetric Area Ratio of Regular Polygons • Cross-Lesson: Topic 5.01 (Ratios) & Topic 8.01 (Polygon Area)

A regular hexagon and an equilateral triangle have the exact same perimeter. What is the ratio of the area of the regular hexagon to the area of the equilateral triangle?

**a.** $1:1$

**b.** $3:2$ ✓

**c.** $2:1$

**d.** $4:3$

Let the common perimeter be $6s$. The hexagon has side $s$ and consists of $6$ equilateral triangles of side $s$. The equilateral triangle has side $2s$ and area equal to $2^2 = 4$ equilateral triangles of side $s$. Thus the area ratio is $6:4 = 3:2$.

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**43** [Challenging] • [MCQ] • Target: Incenter Angle Formula • Cross-Lesson: Topic 2.01 (Linear Angle Systems)

In $\Delta ABC$, angle bisectors of $\angle B$ and $\angle C$ intersect at the incenter $I$. If $\angle A = 68^\circ$, what is the measure of $\angle BIC$?

**a.** $112^\circ$

**b.** $124^\circ$ ✓

**c.** $134^\circ$

**d.** $146^\circ$

Since $\angle B + \angle C = 180^\circ - 68^\circ = 112^\circ$, half their sum is $\frac{1}{2}(\angle B + \angle C) = 56^\circ$. In $\Delta BIC$, $\angle BIC = 180^\circ - 56^\circ = 124^\circ$ (in general, $90^\circ + \frac{1}{2}\angle A$).

**44** [Challenging] • [MCQ] • Target: Crossed-Ladders Similar Triangles Theorem • Cross-Lesson: Topic 10.01 (Harmonic & Rational Equations)

Two poles of heights $10\,\text{m}$ and $15\,\text{m}$ stand vertically on level ground $20\,\text{m}$ apart. A taut wire connects the top of each pole to the base of the opposite pole. At what height above the ground do the two wires intersect?

**a.** $5\,\text{m}$

**b.** $6\,\text{m}$ ✓

**c.** $6.25\,\text{m}$

**d.** $7.5\,\text{m}$

By the crossed-ladders harmonic property (independent of the horizontal separation $20\,\text{m}$), the intersection height $h$ satisfies $\frac{1}{h} = \frac{1}{10} + \frac{1}{15} = \frac{5}{30} = \frac{1}{6} \Rightarrow h = 6\,\text{m}$.

**45** [Challenging] • [MCQ] • Target: Coordinate Triangle Area via Bounding Box / Shoelace • Cross-Lesson: Topic 1.01 (Coordinate Plane)

What is the area of the triangle with vertices $(0, 0)$, $(6, 2)$, and $(3, 8)$ in the $xy$-plane?

**a.** 18

**b.** 21 ✓

**c.** 24

**d.** 27

By the shoelace formula (or bounding box $6 \times 8 = 48$ minus three right triangles of areas $6, 12, 9$), Area $= \frac{1}{2}|6(8) - 2(3)| = \frac{1}{2}(48 - 6) = 21$.

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